G H Raisoni International Skill Tech University, Pune
(Established by Maharashtra Act No. IX of 2024 and under section 2(f) of the UGC Act 1956)
FY B. TECH (ALL) (SEM-II)
ESE SUMMER 2026 (2025 Pattern)
Engineering Physics (25UBSL1204)

[Time:2.5 Hours] **[Max. Marks-60]**

Instructions to the candidates:
1) ***All questions compulsory.***
2) ***Figures to the right indicate full marks.***
3) ***Scientific calculator is allowed.***
4) ***Draw neat diagrams whenever necessary.***
Constants: $c = 3\times10^{8}$ m/s, $h = 6.63\times10^{-34}$ Js, $e = 1.6\times10^{-19}$, $m_e = 9.1\times10^{-31}$ Kg

Q.No.	*Sub Question*	**i) For Q. No. 1, 2,3, sub question (a) is compulsory and sub-question (b) OR (c). ii) For Q. No. 4 and 5 all sub-questions (a, b, c) are compulsory.**	*Marks*	*CO*	*BL*
1	a)	What do you mean by thin film? Derive an expression for Optical path difference due to interference of monochromatic ray of reflected light by thin film of constant thickness. Also deduce condition for constructive and destructive interference.	[5]	CO1	L2
	b)	Explain the construction and working principle of diffraction grating. Determine the condition for principal maxima with well levelled diagram.	[5]	CO1	L1
		OR			
	c)	Using Heisenberg's Uncertainty Principle, show that electron cannot exist within the nucleus.	[5]	CO4	L3
2.	a)	Differentiate between unpolarized light and plane-polarized light. Explain the working principles of a polarizer and an analyzer. State and derive Malus' Law, and discuss its significance in the study of polarization.	[5]	CO2	L2
	b)	With the help of neat labelled diagrams explain the construction and working of Ruby Laser.	[5]	CO3	L1
		OR			
	c)	State the Difference between Metal, insulator and semiconductor on the basis of energy band gap. Give the differences between intrinsic and extrinsic semiconductors.	[5]	CO4	L2
3.	a)	Derive the relationship between the Einstein coefficients for spontaneous emission (A_{21}), stimulated emission (B_{21}), and absorption ($B_{(12)}$). Hence, explain how these coefficients lead to the condition required for laser action.	[5]	CO3	L2

	b)	What is effect of Temperature and doping concentration on Fermi Energy in N type semiconductor? Explain with necessary diagram.	[5]	CO4	L3
		OR			
	c)	Derive the expression for energy Eigen values for a free particle in one dimensional potential well.	[5]	CO4	L2

For Q. No. 4 and 5 all sub-questions (a, b, c) are compulsory.

4.	a)	What is Anti-Reflection Coating? Discuss it in details with phase and amplitude conditions.	[5]	CO1	L3
	b)	Explain the working principle and construction of the semiconductor laser with neat diagram and list the application of laser in medical field.	[5]	CO2	L2
	c)	Explain the formation of interference fringes in Young's Double Slit Experiment. Derive the expression for fringe width and discuss the effect of source separation and slit separation on the pattern.	[5]	CO1	L3
5	a)	Explain the Term: (i) Stimulated absorption (ii) Spontaneous emission (iii) Active medium (iv) Metastable state (v) Life Time	[5]	CO2	L2
	b)	Classify optoelectronic device and explain the working principle and construction of Photo diode with neat diagram.	[5]	CO3	L3
	c)	Define diffraction of light. Write their types. Differentiate between Fresnel diffraction and Fraunhofer diffraction.	[5]	CO2	L2

End of the paper.